

QUANTUM GASES

Realization of three- and four-body interactions between momentum states in a cavity

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Spin Hamiltonians in condensed matter and quantum sensing typically utilize pairwise or two-body interactions between constituents in the material or ensemble. However, there is growing interest in exploring more general n -body interactions for $n > 2$. In this study, we realized an effective $n = 3$ -body Hamiltonian interaction using an ensemble of laser-cooled atoms in a high-finesse optical cavity with the pseudospin $1/2$ encoded by two atomic momentum states. We applied two dressing tones that induce the atoms to exchange photons via the cavity to realize a virtual six-photon process; lower-order interactions destructively interfered. We also observed signatures of a $n = 4$ -body interaction mediated by a virtual eight-photon process. Our approach may be extensible to three-body interactions in multilevel systems or to even higher-order interactions.

Physical systems can often be described by pairwise interactions between objects. However, higher-order interactions, in which three or more objects directly couple, appear in many different contexts. For instance, higher-order interactions have been theoretically predicted to play a key role in understanding complex biological and social networks (1, 2) and in quantum information (3–6), molecular and chemical physics (7, 8), condensed matter (9–13), nuclear physics (14), and high energy physics (15). Multibody interactions cannot be decomposed into a sum of pairwise interactions but instead emerge from effective theories in which intermediate states can be integrated out to consider only dynamics within a restricted Hilbert space. Experimental challenges include not only implementing the n -body interaction in a genuine many-particle system with more than a few atoms but also eliminating the lower-order interactions so that the n -body interactions are dominant.

Regardless of this broad interest, there has been limited progress toward engineering controllable n -body interactions with $n > 2$, with only a few experimental implementations such as superconducting qubits (16–18), trapped ions (19–23), ultracold gases (24–28), and Rydberg atoms (29–31). Scaling of these multibody interactions for quantum simulation and sensing to larger systems is important but difficult. Therefore, their realization in a scalable system such as an optical cavity is highly desirable.

In this study, we experimentally realized a three-body interaction between a thousand rubidium atoms, each prepared in a superposition of two momentum states that act as pseudospin states. The interaction is all-to-all(-to-all) and consists of triplets of atoms all flipping their momentum or pseudospin states at once as shown in Fig. 1A. This is achieved by applying two dressing laser tones along the axis of an optical cavity to realize a resonant three-atom and six-photon process with four of the atom-photon interactions mediated by the cavity mode. By setting the power ratio between the dressing laser tones to one, a pure resonant three-body interaction can be achieved with lower-order two-body interactions (exchange and pair raising and lowering processes)

cancelled through symmetry (32, 33). The collective nature of the cavity-mediated interactions is essential to make the three-body terms comparable to previously demonstrated two-body interactions because the slower rate imposed by the exchange of six photons is compensated by the fact that the strength of the three-body term scales with the atom number as N^2 , compared with the N^1 scaling of the two-body counterpart. We also demonstrate the extension of our approach to n -body with $n > 3$ by observing signatures of a resonant four-body interaction.

Experimental setup

In the experiment, rubidium-87 (^{87}Rb) atoms are laser-cooled inside a vertically oriented standing wave cavity [see Fig. 1B and (32, 34)]. The atoms are allowed to fall along the cavity axis, guided by a repulsive radial potential that is provided by a blue-detuned and hollow optical dipole trap. To prepare and manipulate the atomic momentum states, two-photon velocity-selective Raman or Bragg transitions can be driven by nonresonantly injecting pairs of laser beams with different frequencies into the cavity. After accelerating under gravity for 20 ms, we select around $N = 1000$ atoms with averaged momentum along the cavity axis $p_0 - \hbar k$ and root mean square momentum spread $\sigma_p < 0.1\hbar k$, where $\hbar$ is the reduced Planck constant, the wave number is $k = 2\pi/\lambda$, and the wavelength is $\lambda = 780$ nm. The atomic internal states are then transferred to the ground hyperfine state $|F = 2, m_F = 2\rangle$, using sequential microwave pulses.

We then apply the Bragg lasers to prepare the atoms in a superposition of two wave packets with momenta centered on $p_0 \pm \hbar k$, which we use to define a qubit between states $|\uparrow\rangle \equiv |p_0 + \hbar k\rangle$ and $|\downarrow\rangle \equiv |p_0 - \hbar k\rangle$. To good approximation, we can neglect the spread in momentum and define $\hat{\Psi}_{\uparrow,\downarrow}^\dagger$ and $\hat{\Psi}_{\uparrow,\downarrow}$ as the operators for creating and annihilating an atom in the momentum states $|\uparrow\rangle$ and $|\downarrow\rangle$. The effect of finite momentum spread and the presence of other order momentum states at $p_0 \pm \hbar k$ are studied in (32). The collective ladder operators $\hat{J}_+ = \hat{\Psi}_{\uparrow}^\dagger \hat{\Psi}_{\downarrow}$ and $\hat{J}_- = \hat{\Psi}_{\downarrow}^\dagger \hat{\Psi}_{\uparrow}$ can then be defined for mapping to a collective pseudospin model. Similarly, the collective spin projection operators can be defined as $\hat{J}_x = \frac{\hat{J}_+ + \hat{J}_-}{2}$, $\hat{J}_y = \frac{\hat{J}_+ - \hat{J}_-}{2i}$, and $\hat{J}_z = \frac{\hat{\Psi}_{\uparrow}^\dagger \hat{\Psi}_{\uparrow} - \hat{\Psi}_{\downarrow}^\dagger \hat{\Psi}_{\downarrow}}{2}$. For all collective spin operators, the expectation values are denoted by $J_\alpha = \langle \hat{J}_\alpha \rangle$ with $\alpha \in [x, y, z, +, -]$.

The two Bragg tones can be described by complex amplitudes in the lab frame α_{B1} and α_{B2} such that the differential phase between the Bragg tones accrues as $\psi_B = \arg(\alpha_{B1}^* \alpha_{B2}) = \omega_z t + \phi_B + \frac{1}{2} \delta_r t^2$. Here, $\omega_z = 2kv_0$ is the two-photon Doppler shift for photons moving along the cavity axis at speed $v_0 = p_0/m_a$, where m_a is the mass of the atom. The frequency splitting $\omega_z/2\pi$ is typically about 500 kHz for our experiments, but because the atoms are accelerating owing to gravity, we linearly ramp the frequency difference of the two tones at a rate $\delta_r = 2\pi \times 25.11$ kHz/ms to compensate. The differential phase ϕ_B sets the azimuthal phase of the Bragg rotation on the Bloch sphere.

Three-body interactions with two-tone dressing

As shown in Fig. 2A, a cavity mode is stabilized to the blue of the D2 cycling transition $|F = 2, m_F = 2\rangle \rightarrow |F' = 3, m_{F'} = 3\rangle$ with a detuning $\Delta_a = \omega_c' - \omega_a = 2\pi \times 500$ MHz. The detuning Δ_a is large compared with all other relevant frequencies, including the excited state decay rate $\Gamma = 2\pi \times 6$ MHz and the cavity power decay rate $\kappa = 2\pi \times 56(3)$ kHz. In this far-detuned limit, an atom at position Z shifts the cavity resonance by $\frac{\kappa_0^2}{\Delta_a} \cos^2(kZ)$, where $2g_0 = 2\pi \times 0.96$ MHz is the maximal single photon Rabi coupling at a cavity antinode (35).

We create the three-body interaction by injecting two dressing laser tones at frequencies ω_1 and ω_2 into the cavity, as shown by the solid red and blue arrows in Fig. 1, B and C. The two tones are separated in frequency by $\omega_2 - \omega_1 = 3\omega_z$, and their amplitudes are described by complex amplitudes α_1 and α_2 , as shown in Fig. 2A.

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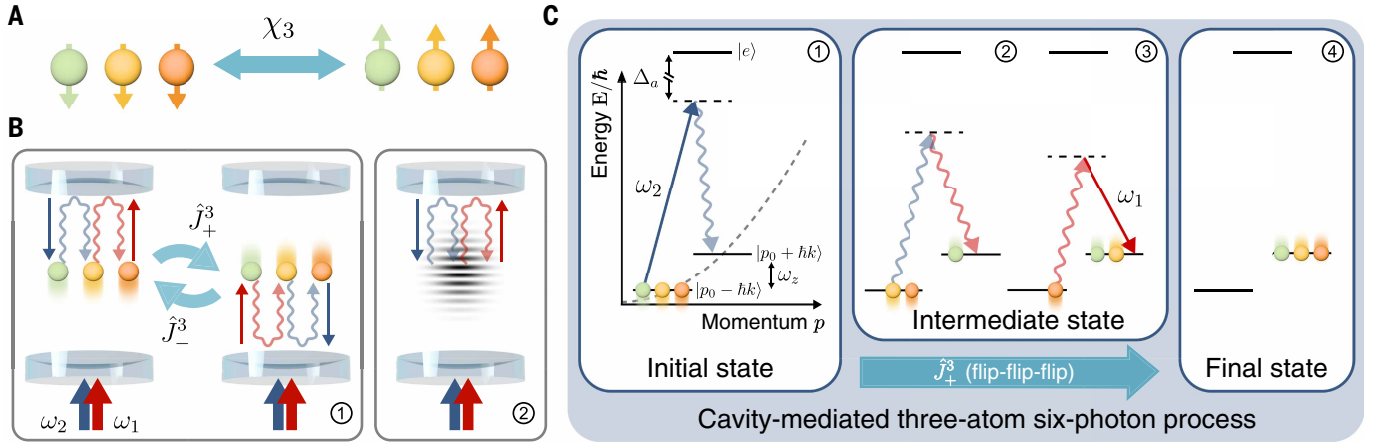


Fig. 1. Three-body interaction. (A) A three-body lowering and raising interaction causes a trio of spins to flip their spin states simultaneously. (B) Here, the pseudospin is a pair of atomic momentum states, $p_0 \pm \hbar k$, along the cavity, separated by two photon-recoil momenta $2\hbar k$. (Left) The momentum relative to the average momentum p_0 is indicated by the tail on each colored atom. The cavity is driven by two dressing lasers (red and blue arrows). The all-to-all three-body interaction is realized by the atoms exchanging photons via the cavity, with the intermediate photons of different frequencies shown (red and blue squiggly arrows). (Right) An atomic density grating (gray) is generated by placing the atoms in a superposition of momentum states, inducing multiple back-reflections of light that lead to the three-body interaction. (C) The energy level diagram for the three-atom and six-photon process illustrates how three atoms flip their momentum states by showing the intermediate virtual states. Only the relevant fields are shown at each step for simplicity. The kinetic energy difference of the two momentum states is $\hbar\omega_z$, and the pumps are detuned by approximately Δ_a from $|e\rangle$. We operate in detuned regimes such that both the intermediate optical excited state and the intermediate photon states (squiggly arrows) can be adiabatically eliminated to arrive at the three-body pseudospin interaction.

The mechanism by which the three-body interaction arises is illustrated in Fig. 1, B and C. A blue dressing laser photon at frequency ω_2 is absorbed and reemitted into the cavity (lighter-blue squiggly arrow) in the opposite direction with a frequency shift $-\omega_z$, as shown in Fig. 1, B1 and C1, and Fig. 2A. This virtual two-photon process transfers $2\hbar k$ of momentum to the absorbing atom, flipping its pseudospin. This process then repeats twice more but with the frequency-shifted photons being absorbed and backscattered by other atoms, at each step receiving a frequency shift $-\omega_z$. This virtual process terminates with stimulated emission into the red dressing laser at frequency $\omega_1 = \omega_2 - 3\omega_z$, and with three different atoms having flipped their momentum or pseudospin state. Although we illustrated the process with three atoms, we note that each step is collectively enhanced by constructive interference between paths in which different atoms in the N atom ensemble undergo the momentum flip, as compared with, for instance, a single atom decreasing its momentum state three times in a row. This constructive interference causes the collective interaction characteristic energy scale to increase as N^2 .

After adiabatic elimination of the intermediate virtual states in the above, this cavity-mediated process manifests as a collective three-body raising and lowering process described by the effective Hamiltonian:

$$\hat{H}/\hbar = \chi_3 \hat{J}_+^3 e^{3i\phi_d} + \chi_3 \hat{J}_-^3 e^{-3i\phi_d} \quad (1)$$

The phase of the interaction $3\phi_d$ is given by the differential phase of the two dressing tones with respect to the Bragg pulse that creates the initial superposition of momentum states $3\phi_d = \arg(\alpha_1^* \alpha_2) - 3\phi_B$ with ϕ_B with phase of the Bragg coupling [see (36) for details]. The interaction strength is given by

$$\chi_3 = \left(\frac{g_0^2}{4\Delta_a} \right)^3 |\alpha_1 \alpha_2| \text{Re} \left[\frac{1}{(\Delta_{c1} + i\kappa/2)(\Delta_{c2} + i\kappa/2)} \right] \quad (2)$$

where the detuning of the red and blue sideband tones (intermediate squiggly arrows) from resonance with the cavity appear in the denominator as $\Delta_{c2} = (\omega_2 - \omega_z) - \omega'_c$ and $\Delta_{c1} = (\omega_1 + \omega_z) - \omega'_c$. Here, $\omega'_c = \omega_c + Ng_0^2/(2\Delta_a)$

is the time-averaged cavity resonance frequency. The incident dressing laser power on the cavity is typically only a few hundred picowatts.

Each dressing tone individually also induces a two-body exchange interaction of the form $\chi_2 \hat{J}_+ \hat{J}_- (33)$. For typical operating conditions we expect $|\chi_3/\chi_2| \ll 1$. However, the collective interaction strengths are in fact comparable $|\frac{\chi_3 N^2}{\chi_2 N}| \approx 1/2$. Moreover, the two-body exchange interactions of the two dressing lasers can be tuned to have opposite signs and cancel by simply operating in the symmetric configuration shown in Fig. 2A, with $\Delta_{c2} = -\Delta_{c1}$ and $|\alpha_2| = |\alpha_1|$. The symmetry thus allows the three-body interactions here to dominate the observed dynamics.

A complementary way to understand the three-atom and six-photon process is in terms of the atomic spatial density grating shown in Fig. 1, B2. The atomic density grating is the modulated quantum probability density of finding the atoms at location Z along the cavity axis, owing to having prepared the atoms in a superposition of momentum states. The density grating has a periodicity of $\lambda/2$ and appears to the light as a spatially varying index of reflection that can reflect the light, much like a dielectric stack mirror. With the atomic density grating shown in Fig. 1, B2, moving along the cavity at velocity v_0 , a downwards-going blue dressing laser photon (solid blue arrow) is reflected by the atomic density grating. The reflected light experiences a Doppler shift of precisely $-\omega_z$, which creates the sideband photon (lighter-blue squiggly arrow), and at the same time an atom receives a two-photon momentum kick $2\hbar k$, flipping its pseudospin state. The sideband photon is then reflected and frequency shifted by the atomic density grating twice more, flipping the momentum or pseudospin state of two other atoms, before the photon is stimulated back into the red dressing laser (solid red arrow).

Mean-field dynamics

Assuming $\phi_d = 0$, the mean-field equations of motion for the collective spin operators can be derived from the effective three-body interaction Hamiltonian along with Lindblad operators that capture collective decay through the cavity (36):

$$\begin{aligned} \dot{J}_+ &= -6i\chi_3 J_z J_-^2 - \Gamma J_+ \\ \dot{J}_z &= -3i\chi_3 (J_+^3 - J_-^3) - 2\Gamma J_z \end{aligned} \quad (3)$$

Here, Γ is the collective or super-radiant decay rate, with the ratio to the interaction strength given by $\Gamma/\chi_3 \sim \frac{4\Delta_a\kappa}{g_0^2}$. In the symmetric configuration used in the experiment, the super-radiance is balanced, resulting in no collective enhancement for this process (36). However, the unitary dynamics driven by the three-body interaction are enhanced by a factor of N^2 (see below) so that $\Gamma/N^2\chi_3 \ll 1$, and to good approximation we can set $\Gamma = 0$ in the subsequent discussion of collective mean-field dynamics. For extending the discussion to $\phi_d \neq 0$, one can simply apply the substitution $J_{\pm} \rightarrow J_{\pm}e^{\pm i\phi_d}$, $J_z \rightarrow J_z$ to Eq. 3.

Figure 2B shows the simulated mean-field dynamics induced by the three-body interactions. The displayed flow lines (gray lines) indicate the trajectory that a Bloch vector follows on the sphere with time (11, 33, 37). Specifically, the local flow vector is defined by $(\mathbf{J}_f - \mathbf{J}_i)/\Delta t$ for any initial Bloch vector $\mathbf{J}_i/(N/2) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ prepared on the Bloch sphere, and with $\mathbf{J}_f$ the final Bloch vector after interacting for a short time Δt . The resulting flow lines predict eight fixed points on the Bloch sphere: two unstable fixed points located at

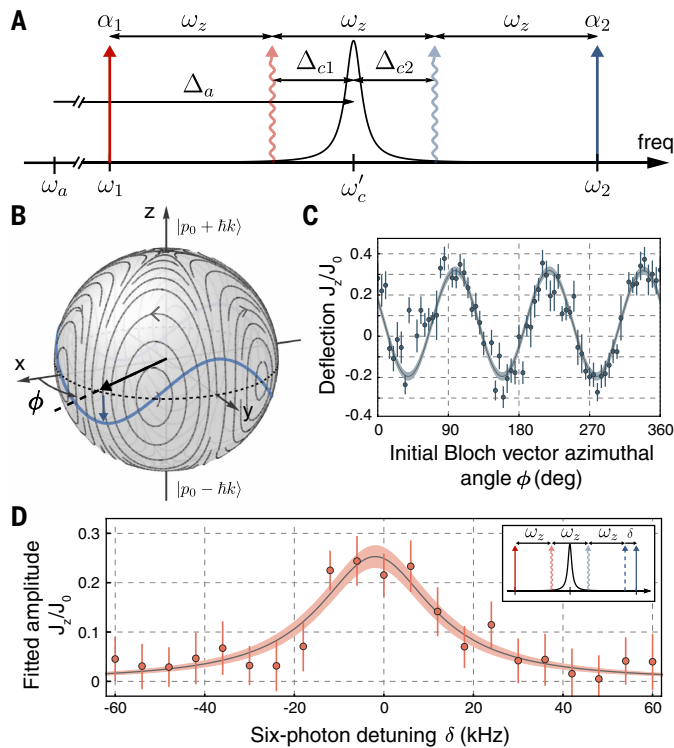


Fig. 2. Dynamics and spectroscopic signal for three-body interaction. (A) Frequency diagram. The cavity mode is detuned by Δ_a from an atomic cycling transition. Two dressing lasers (red and blue arrows) are applied at frequencies $\omega_{2,1}$, with frequency difference $3\omega_z$, and amplitudes $\alpha_{2,1}$. Scattering from the atoms generates the two new sideband tones (red and blue squiggly lines). The four tones combine to resonantly drive the three-atom and six-photon process illustrated in Fig. 1. (B) Numerically simulated mean-field dynamics (36) induced by the three-body interaction. The flow lines indicate the direction that a Bloch vector prepared at a given point on the Bloch sphere will subsequently move with time. The blue curve is the predicted short time displacement of the spin projection J_z for Bloch vectors initially prepared at different azimuthal angles on the equator ($\theta = \pi/2$, dashed black line.) (C) Experimentally measured displacements of J_z owing to the three-body interaction, normalized by $J_0 = N/2$. The data and fit display the predicted three-cycle oscillation as the initial azimuthal angle ϕ is varied from 0 to 360° . (D) The 3-cycle oscillation of (C) is observed only when the six-photon process is resonant. The fitted amplitude of the 3-cycle oscillation is shown as a function of the six-photon detuning $\delta = \omega_1 - \omega_2 - 3\omega_z$, see frequency diagram in inset. Data are fit with 68% confidence bands and all error bars reported are 1σ uncertainties.

the north and south poles with $\mathbf{J}_i/(N/2) = (0, 0, \pm 1)$ (i.e., $\theta = 0, \pi$) and six stable fixed points located on the equator (i.e., $\theta = \pi/2$) and evenly spaced in their azimuthal angles ϕ by $\pi/3$. These are consistent with the Hamiltonian being symmetric under a $\pi/6$ rotation around $\hat{z}$ and a time-reversal operation.

In Fig. 1B we consider an ensemble of initial states prepared on the equator ($\theta = \pi/2$, black dashed line) with different azimuthal angles chosen in the range $\phi \in (0, 2\pi)$. The blue curve shows the predicted change in spin projection $\Delta J_z/(N/2) = \frac{3}{2}\chi_3 N^2 \Delta t \sin(3\phi)$ along $\hat{z}$ after the short-time evolution, which reflects the N^2 enhancement factor. When the initial Bloch vector is prepared at a stable fixed point, there is no change in the Bloch vector orientation $\Delta \mathbf{J} = 0$, here occurring at azimuthal angles $\phi = q\pi/3$ with q an integer.

To understand the unstable fixed points, we project the theoretical flow vector field onto the $\hat{x} - \hat{y}$ plane (gray arrows in Fig. 3A) as viewed looking down at the north pole of the Bloch sphere. Several key directions are highlighted below. From Eq. 3, we find $\dot{J}_z = 0$ at $\phi = q\pi/3$, meaning that the flow lines exhibit only an azimuthal component at these points. Furthermore, the equation for $\dot{J}_+$ can be reformulated in terms of (θ, ϕ) , where $\phi = -\frac{3}{4}\chi_3 N^2 \cos(3\phi) \sin(2\theta)$, implying that the azimuthal angle does not change at initial Bloch vector angles $\phi = \pi/6 + q\pi/3$, and only the polar angle changes at these azimuthal angles. Additionally, the expression for $\dot{\theta} = -\frac{3}{2}\chi_3 N^2 \sin^2\theta \sin(3\phi)$ shows that $\dot{\theta} > 0$ for $\phi = \pi/6 + 2q\pi/3$ and $\dot{\theta} < 0$ for $\phi = \pi/2 + 2q\pi/3$. These features define the trifurcation points at the north and south poles;

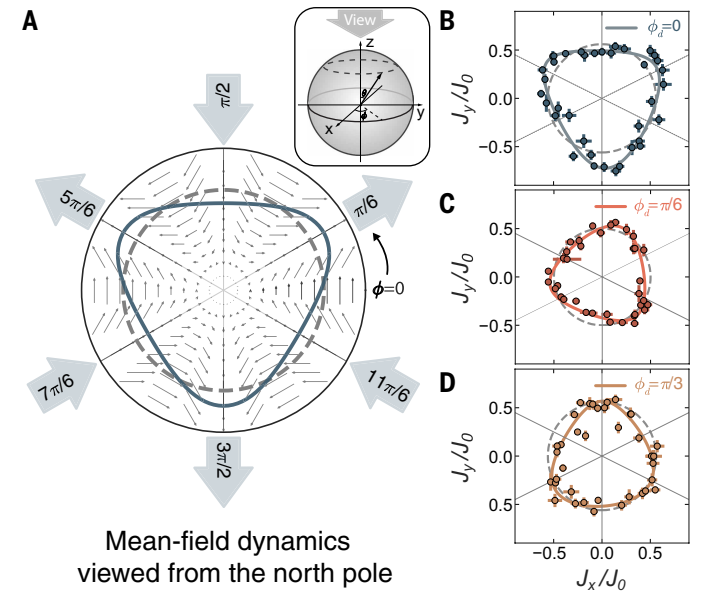


Fig. 3. Deformation of circular distribution. (A) Theoretical mean-field flow lines shown as a vector field (light gray arrows) and as viewed from above the north pole (see inset.) The thick arrows surrounding the Bloch sphere highlight the fact that near the trifurcation points, the flow lines alternate every $\pi/3$ in azimuthal angle between flowing toward versus away from the unstable trifurcation point at the north pole. As a consequence, after a short time of evolution, a distribution of Bloch vectors prepared at different ϕ but constant polar angle $\theta = \pi/4$, illustrated by the dashed gray circle, will evolve for short interaction times into a distribution shown by the dark-blue curve. (B) Experimental observation of the predicted mean-field dynamics in (A). The distribution of final measured Bloch vectors on the $J_x - J_y$ plane deforms away from a circular distribution after the three-body interaction is applied for a short time. (C and D) The final Bloch vector distribution is rotated by $\pi/6$ and $\pi/3$ by repeating the experiment but changing the relative phase $3\phi_d$ between the two dressing tones by $\pi/2$ and π , respectively. This shows that the orientation of the three-body interaction can be controlled through the dressing laser phases.

e.g., near the north pole trirforkation point, the flow lines alternate every $\pi/3$ in azimuthal angle between flowing toward ($\dot{\theta} < 0$) versus away from ($\dot{\theta} > 0$) the trirforkation point, as shown by the thick arrows in Fig. 3A.

Observing dynamics near the stable fixed points

To experimentally observe the predicted dynamics, we apply a Bragg $\pi/2$ -pulse lasting 30 μs to prepare the initial Bloch vector on the equator (i.e., $J_z = 0$). After 50 μs of separation, a Bragg π -pulse is applied to refocus the wave packets. We applied the dressing lasers for 50 μs after the wave packets re-overlap with $\chi_3 N^2 = 2\pi \times 390$ Hz. The population in each momentum state is measured at the end to determine the spin projection J_z . We then repeat the measurement sequence for different initial Bloch vector azimuthal angles ϕ , achieved by changing the Bragg pulse phase ϕ_B while holding the dressing laser phase $3\phi_d$ fixed. Figure 2C shows that the measured spin projection J_z oscillates up and down three times as ϕ_B , and thus ϕ is varied 0 to 2π . The six azimuthal angles at which maximum deflections occur are points half-way between adjacent fixed points where the quasircular flows move in the same directions. The six points between maximum and minimum deflection occur at azimuthal angles that are aligned to the stable fixed points.

We also experimentally confirm that the measured deflections of the spin projection J_z arise from the resonant six-photon interaction outlined in Fig. 1. To do this, we move the two dressing lasers away from the six-photon resonance condition by setting $\omega_2 - \omega_1 = 3\omega_z + \delta$. The frequency ω_z is independently determined via Bragg spectroscopy. (For simplicity in this discussion, we neglect the frequency chirping at rate δ_r that is still applied as previously described.) We then scan the Bloch vector's azimuthal angle 0 to 2π and extract the three-cycle fringe height at a given detuning δ . As shown in Fig. 2D, one sees a resonance in the fringe height near $\delta = 0$ as expected and with a full width at half maximum of 28 kHz, determined by the duration of the interaction.

Observing dynamics near an unstable fixed point

We experimentally measure the dynamics near the unstable trirforkation point at the north pole by observing the evolution of Bloch vectors prepared at initial azimuthal angles $\phi = 0$ to 2π but at a constant polar angle $\theta = \pi/4$ (chosen to maximize the visibility of the mean-field trirforkation dynamics). To prepare these initial states, we start with the Bloch vector at the south pole. We then apply an initial Bragg $\pi/4$ -pulse. After 50 μs , we apply a Bragg π -pulse that refocuses the matter-wave packets, while also setting the polar angle to $\theta = \pi/4$. The interaction is then applied when the two wave packets are maximally re-overlapped for 50 μs . A final Bragg $\pi/2$ -pulse is applied along $\hat{y}$ or $\hat{x}$ to map the spin projections J_x or J_y onto J_z , the spin projection which we can measure (36).

Without the interaction applied, we expect a circular distribution of final Bloch vectors centered around $\hat{z}$. After evolution under the three-body interaction, the measured circular distribution shown in Fig. 3B is seen to deform into a more triangular distribution, which is consistent with the predicted trirforkation point at the north pole, where there are three converging flow lines equally interspersed between three diverging flow lines. We also demonstrate phase control of the interaction by repeating the experiment but changing the phase of the interaction $\phi_d \rightarrow \phi_d + \pi/6$ and $\phi_d + \pi/3$. In Fig. 3, C and D, one sees that the deformation of the originally circular distribution is also rotated by $\pi/6$ and $\pi/3$, respectively.

Signatures of a four-body interaction

We also realize a four-body interaction of the form $\hat{H}_4/\hbar = \chi_4^+ \hat{J}_+^4 e^{4i\phi} + \chi_4^- \hat{J}_-^4 e^{-4i\phi}$ and observe a signature of the expected dynamics. To achieve this, we apply the dressing lasers with a frequency difference $\omega_2 - \omega_1 = 4\omega_z$ (Fig. 4A), which brings a four-atom and eight-photon process (Fig. 4B) into resonance. The two-body exchange interactions can again be

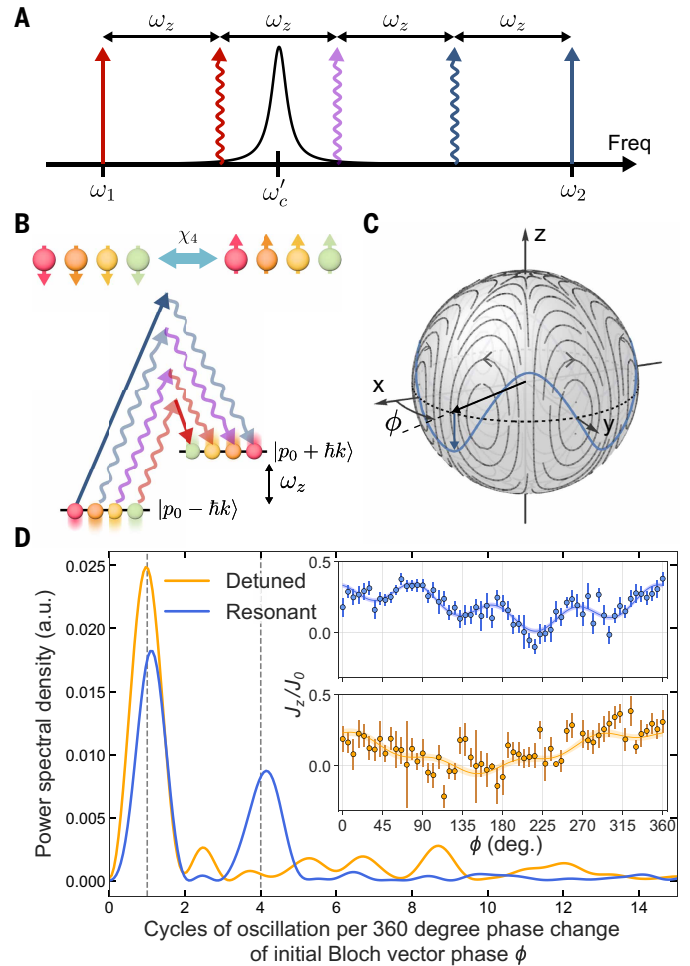


Fig. 4. Signature of four-body interaction. (A) Frequency diagram for driving four-body interaction. The two dressing laser tones (red and blue arrows) are separated by $4\omega_z$, and the intermediate photons (squiggly lines) are equally separated by ω_z . (B) Energy-level diagram for the four-atom and eight-photon process that flips the momentum state of four items simultaneously. (C) Theoretical flowlines for the mean-field dynamics. With four stable points on the equator, the Bloch vector will experience a 4-cycle oscillation in the change of the spin projection J_z when scanning the azimuthal phase of the initial Bloch vector. (D) (Insets) The measured deflection J_z caused by applying the four-body interaction versus the azimuthal phase of the initial Bloch vector. Blue data and fit are for the dressing lasers on resonance for the eight-photon process. Yellow data and fit are for the dressing lasers tuned away from resonance. A 4-cycle oscillation of J_z can be discerned in the resonant case. (Main panel) The power spectrum of the data shows a 4-cycle peak at resonance, in contrast to off-resonance. The 1-cycle peak appears for both on- and off-resonance data with a similar amplitude for both.

cancelled by adjusting the relative power in the dressing laser, and the three-body interaction is nonresonant in this case. The characteristic frequency scale of this interaction is now collectively enhanced as N^3 , but the larger detuning from cavity resonance of the additional intermediate photon leads to a smaller predicted interaction strength $\chi_4 N^3 \approx 2\pi \times 130$ Hz. The predicted mean-field dynamics are illustrated by the calculated flowlines shown in Fig. 4C.

To observe dynamics induced by this four-body interaction, we again prepare the initial Bloch vector on the equator with $\theta = \pi/2$ and measure the change in J_z versus initial Bloch vector azimuthal phase ϕ . The measured fringe is shown in Fig. 4D, inset, with the two dressing lasers tuned to resonance (blue) and detuned from resonance (yellow).

One can see a clear 4-cycle oscillation in the displacement of J_z as expected from the eight stable points on the equator that result from a four-body interaction. The main panel shows the Fourier power spectra of the insets with a clear peak at the 4-cycle frequency occurring above the noise floor only when the four-photon coupling is on resonance. The data were zero-padded before the Fourier transform to interpolate the spectrum to a finer frequency resolution for visualization; this does not alter the underlying spectral content. Although not included here, the Fourier spectrum of the oscillation induced by the three-body interaction in Fig. 2C produces a peak centered at the 3-cycle frequency.

There is also a 1-cycle Fourier component that is apparent in the inset traces of J_z versus ϕ . The blue and yellow solid lines in Fig. 4D are fitted to a sum of 1- and 4-cycle oscillations with a function $y_0 + A_1 \sin(\phi - \phi_1) + A_4 \sin(4\phi - \phi_4)$. The similar amplitude of the 1-cycle component in the eight-photon resonant versus nonresonant data indicates that this is not related to a resonant interaction between the dressing tones, and n -body interactions that depend on a single dressing tone should not depend on the phase of the dressing light. We currently attribute this spectral feature to single-particle physics caused by an unwanted Bragg coupling driven by a spurious sideband that appears in the generation of the two dressing laser tone.

Discussion and outlook

Our platform can allow for more energy levels and more atoms [limited here by needing to do narrow velocity selection from a larger initial thermal ensemble (32, 33)], allowing for full connectivity, which would also be worthwhile to combine with local control in the future (38–40). Although demonstrated using momentum states in a cavity, the interaction can also be applied using internal atomic or molecular energy levels.

The technique demonstrated here provides a distinct set of tools for quantum simulation and entanglement generation using higher-order correlations. This opens the path to a broad host of future explorations of exotic many-body physics, from new regimes of self-organization physics (41, 42) to quantum simulation of charge-4e superconductors (13), as well as quantum metrology with new classes of entangled states. It will also be beneficial to understand how single-particle and collective dissipation (32, 43) (predominantly described by jump operators $\hat{J}_\alpha$) compete with entanglement arising from the n -body Hamiltonian dynamics. Furthermore, even though we have discussed the implementation of these schemes in optical cavities, our protocols can be straightforwardly generalized to other types of systems where interactions are mediated by a bosonic mode such as trapped ions or superconducting qubits. Because different platforms are affected by different types of decoherence processes, it will be useful to understand optimal performance in the different settings.

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SUPPLEMENTARY MATERIALS

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Realization of three- and four-body interactions between momentum states in a cavity

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Editor's summary

Interactions between pairs of particles often dominate over those in which three or more particles couple directly. To implement such higher-order interactions experimentally, pairwise interactions need to be suppressed. Luo *et al.* realized three- and four-body interactions between 1000 rubidium atoms held in an optical cavity. Lower-order two-body interactions were cancelled through symmetry. This technique may enable future explorations of exotic many-body physics as well as advances in quantum metrology. —Jelena Stajic

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